

CONFIDENTIAL — ACADEMIC SUBMISSION

DIGITAL FORENSICS & INCIDENT RESPONSE

FORENSIC INVESTIGATION REPORT — FINAL EDITION

*Unknown Breach: Evidence Discovery, Validation,
and Full Forensic Investigation*

Field	Value
Case Title	M57.Biz — Corporate Data Exfiltration Incident
Case Number	DFIR-2024-012
Evidence Image	jo-2009-11-16.E01
Computer Name	M57-JO (Windows XP SP3, x86)
Subject	Jo — Employee, M57.biz
Examiner	Youssef Moataz — AASTMT Cybersecurity, Class of 2026
Course	Digital Forensics & Incident Response (CCY3102)
Institution	Arab Academy for Science, Technology & Maritime Transport
Date Submitted	April 2026
Primary Tools	Autopsy 4.23 FTK Imager 4.7.3.81 Velociraptor (simulated)
Verified MD5	f3160a776ca1d59172c6acb622b67459 ✓ MATCH
Verified SHA1	cf2d8ca1e282501dc7977acd4ac48df1a4b23e9f ✓ MATCH
Report Version	v2.0 — Final Edition (all artefact layers integrated)

This report is prepared solely for academic evaluation purposes.

Executive Summary

This final edition forensic investigation was conducted against **jo-2009-11-16.E01**, sourced from the M57-Patents dataset (Digital Corpora, UCSD). The machine **M57-JO** ran Windows XP SP3 and was registered to employee **Jo** of **M57.biz**. The image was captured on 16–17 November 2009.

This edition incorporates analysis of **Event Logs (Application + System)**, **Registry hives (SYSTEM, SOFTWARE, NTUSER.DAT)**, **Prefetch files (86 entries including unallocated residue)**, **keyword search results**, **web form autofill data**, **remote-monitoring artefacts**, and metadata. A critical new finding — **ZIPPER.EXE in unallocated Prefetch space** — elevates the exfiltration assessment significantly.

[CRITICAL FINDING — UPGRADED] ZIPPER.EXE (file compression tool) was identified in unallocated Prefetch space, indicating it was previously executed and the Prefetch entry was subsequently overwritten — a hallmark of deliberate cleanup. Combined with the LaCie HDD connection, IMAPI CD-burning activity confirmed via System Event Log (Event 7036 — 2009-11-17 02:31:53), and HELIX.EXE execution immediately followed by an Event Log shutdown (6006 at 05:08:26), the evidence profile is now consistent with premeditated data compression, exfiltration, and partial anti-forensic cleanup.

Finding	Severity	Evidence Source	Status
ZIPPER.EXE in unallocated Prefetch — file compression tool previously run	CRITICAL	Prefetch_20260429123907.csv	CONFIRMED
LaCie Rugged HDD connected at 02:20:07 on acquisition day	CRITICAL	USB_Device_Attached CSV	CONFIRMED
IMAPI CD-Burning Service: RUNNING → STOPPED on 2009-11-17 02:31:53–59 (Event 7036)	CRITICAL	sys-event.xlsx	CONFIRMED
Fastfat Warning (Event 50) on \Device\HarddiskVolume3 at 02:25:28 — removable volume	HIGH	sys-event.xlsx	CONFIRMED
HELIX.EXE run at 05:07:23 → EventLog stopped at 05:08:26 (1 min later)	HIGH	Prefetch + sys-event.xlsx	CONFIRMED
MDD_1.3.EXE (memory dump) executed at 02:22:34 on acquisition day	HIGH	Prefetch CSV	CONFIRMED
webmail.m57.biz: login_username='jo' submitted 5 times — failed login confirmed	HIGH	Web_Form_Autofill CSV	CONFIRMED
USPTO.gov patent office browsed extensively (16 pages) before imaging	MEDIUM	Web_History CSV	CONFIRMED

Finding	Severity	Evidence Source	Status
mstsc.exe (Remote Desktop Client) flagged as notable — possible lateral movement	MEDIUM	Remote_Monitoring CSV	CONFIRMED
DEFRAG.EXE executed 2009-11-15 — possible anti-forensic activity	MEDIUM	Prefetch CSV	CONFIRMED
No malicious persistence in Run keys — insider threat confirmed	INFO	Registry / Autopsy	CONFIRMED
Keyword search: no literal/regex hits — no plaintext exfil strings in allocated space	INFO	TSK_KEYWORD_HIT CSV	CONFIRMED

Exfiltration Assessment: Based on the totality of evidence — ZIPPER.EXE execution (now deleted), LaCie HDD connection, IMAPI CD-burning confirmed by System Event Log, and HELIX.EXE counter-forensic activity — data exfiltration is assessed as **HIGHLY PROBABLE** with medium-high confidence. The specific channel (USB, optical disc, or webmail) cannot be conclusively determined from disk evidence alone.

Step 0 — Evidence Sourcing & Dataset Justification

0.1 Dataset Identification

Attribute	Details
Dataset Name	M57-Patents Scenario
Repository	Digital Corpora (digitalcorpora.org) — University of California San Diego
Primary Image	jo-2009-11-16.E01 (Jo's workstation)
Image Format	Expert Witness Format (E01) — EnCase-compatible multi-segment
OS in Image	Microsoft Windows XP Service Pack 3 (x86)
Computer Name	M57-JO
Registered Owner	Jo / Organisation: M57.biz / Product ID: 76487-027-5250835-22087
Acquisition Date	16–17 November 2009
Image Path Used	G:\proj\jo-2009-11-16.E01
Autopsy Case	DFIR_Project (jo-2009-11-16.E01_1 Host)
License	Creative Commons — open for academic and research use

0.2 Justification for Selection

- **Realism:** Created by Garfinkel et al. (UCSD) to simulate genuine corporate activity — real user browsing, email, and file system operations, not synthetic artefacts.
- **Scenario Alignment:** The M57 scenario depicts a corporate environment with a suspected insider data leak, directly matching the project brief.
- **Artefact Richness:** Contains Prefetch files (86 entries), Registry hives (SYSTEM, SOFTWARE, SAM, SECURITY, NTUSER.DAT), Event Logs (AppEvent.Evt, SysEvent.Evt, SecEvent.Evt), browser history, email client data, USB entries — all 9 required analysis domains covered.
- **Tool Compatibility:** E01 format natively supported by FTK Imager 4.7.3.81 and Autopsy 4.23 with no conversion.
- **Verified Provenance:** Hosted by Digital Corpora (UCSD) with documented chain of custody. Cited in peer-reviewed DFIR literature and used in university curricula worldwide.

0.3 Dataset Limitations

- **LaCie Drive Not Imaged:** The LaCie Rugged HDD connected before acquisition is the most critical physical evidence item but was not included in the E01 image. Its content remains unknown from this investigation alone.
- **No RAM Image Available for Analysis:** Although MDD_1.3.EXE was run and a /RAM directory is referenced in Prefetch, the resulting memory dump is not part of the E01. Volatile state (running processes, network connections, decryption keys) is unavailable.
- **No Network Capture:** PCAP/NetFlow data is unavailable. Web-based upload channels (HTTPS, webmail) cannot be confirmed or denied from disk alone.

- **ShimCache Not Parsed:** The SYSTEM hive was confirmed present (3.4 MB, last modified 2009-11-17 02:22:31 EET). ShimCache parsing was not completed due to tooling limitations — see Task 4.5.
- **Amcache Not Applicable:** Windows XP SP3 predates the Amcache artefact (introduced Windows 8). Not applicable — see Task 4.6.
- **Security Event Log Sparse:** SecEvent.Evt was captured but contained no entries post-2009-11-08 — it was cleared or overwritten during initial setup, providing limited logon/object access data.
- **2009 XP SP3 Context:** Artefact locations differ from modern Windows. Notable absences include Jump Lists (Win7+) and VSS Shadow Copies (not detected).

Task 1 — Evidence Validation

1.1 Tool: FTK Imager 4.7.3.81

The E01 image was loaded into FTK Imager (Exterro v4.7.3.81). FTK reads acquisition hashes from E01 case metadata and recomputes them sector-by-sector for verification.

1.2 Hash Verification Results

Algorithm	Stored Hash (Acquisition)	Computed Hash (Verification)	Result
MD5	f3160a776ca1d59172c6acb622b67459	f3160a776ca1d59172c6acb622b67459	✓ MATCH
SHA1	cf2d8ca1e282501dc7977acd4ac48df1a4b23e9f	cf2d8ca1e282501dc7977acd4ac48df1a4b23e9f	✓ MATCH

```
FTK Imager v4.7.3.81 — Drive/Image Verify Results
-----
Name           : jo-2009-11-16.E01
Sector count   : 25,429,824
MD5 → Computed : f3160a776ca1d59172c6acb622b67459 [ MATCH ✓ ]
SHA1 → Computed : cf2d8ca1e282501dc7977acd4ac48df1a4b23e9f
Bad blocks     : No bad blocks found in image [ CLEAN ✓ ]
INTEGRITY RESULT : VERIFIED — Image is forensically sound.
```

1.3 Registry Hive Identification

Registry hives confirmed present in C:\WINDOWS\system32\config\ (source: config_20260428115929.csv):

Hive File	Size	Last Modified (EET)	Contents
system	3.4 MB	2009-11-17 02:22:31	SYSTEM hive — USBSTOR, services, ShimCache (AppCompatCache)
software	11.5 MB	2009-11-17 02:32:38	SOFTWARE hive — installed programs, Run keys
SAM	256 KB	2009-11-17 02:32:38	SAM — user account database
SECURITY	256 KB	2009-11-17 02:32:38	SECURITY — policy, audit settings
AppEvent.Evt	64 KB	2009-11-17 02:32:16	Application Event Log — 91 events
SysEvent.Evt	128 KB	2009-11-17 02:32:16	System Event Log — 169 events
SecEvent.Evt	64 KB	2009-11-08 17:53:06	Security Event Log — minimal (cleared during setup)
NTUSER.DAT	N/A	N/A	Jo's user hive — UserAssist, shell bags, MRU

[FINDING 1.1] Hash verification confirms the forensic image is intact and unmodified since acquisition. All registry hives and event logs are present and accessible. Chain of custody is established.

Task 2 — Initial Triage

2.1 Operating System Identification

Parameter	Value
OS Name	Microsoft Windows XP Service Pack 3
Architecture	x86 (32-bit)
Computer Name	M57-JO
Registered Owner	Jo
Organisation	M57.biz
Windows Path	C:\WINDOWS
Temp Directory	%SystemRoot%\TEMP
Product ID	76487-027-5250835-22087
First Boot	2009-11-08 17:38:17 EET (EventLog first started — MACHINENAME)
Last EventLog Stop	2009-11-17 02:32:15 EET (Event 6006 — system shutdown/imaging)

2.2 User Accounts

Username	Type	Profile Path	Notes
Jo	Standard User	C:\Documents and Settings\Jo\	Primary subject — all user-level artefacts
Administrator	Administrator	C:\Documents and Settings\Administrator\	Used for system setup: AVG, Java, OpenOffice installs

2.3 Key Installed Software (Chronological)

Program	Install Date (EET)	Forensic Relevance
AVG 9.0	2009-11-09 01:14:58	Antivirus — active throughout; AV logs may contain file scan history
OpenOffice.org 3.1	2009-11-10 01:01:42	Document editor — Definitions.odt recently accessed
Java 6 Update 17	2009-11-10 03:31:48	Confirmed installed via AppEvent 11707 — JRE for Python/OO
Mozilla Firefox 3.5.5	2009-11-13 01:52:06	Primary browser — installed on day of HELIX activity
Python 2.6.4	2009-11-13 03:59:27	Confirmed via AppEvent 11707 — scripting runtime

Program	Install Date (EET)	Forensic Relevance
WebFldrs XP	2009-11-11 02:12:54	Confirmed via AppEvent 11728 — web folder client
mstsc.exe	Built-in XP	Remote Desktop Client — flagged as notable; accessed 2009-11-11

2.4 Autopsy Artefact Summary

Category	Count	Priority
Web History (Firefox + IE)	3,557	● HIGH
Web Bookmarks	96	○ MEDIUM
Web Cookies	309	○ LOW
Web Search	36	● HIGH
Web Downloads	3	● HIGH
Web Form Autofill	8	● HIGH — login_username='jo' × 5
USB Device Attached	6	● CRITICAL
E-Mail Messages	2	○ MEDIUM — old test msgs only
Run Programs (Prefetch)	49	● HIGH
Prefetch Files (raw)	86	● HIGH — includes unallocated entries
Shell Bags	25	○ MEDIUM
Recent Documents	2	○ MEDIUM
Installed Programs	95	○ MEDIUM
Extension Mismatch	94	○ MEDIUM
Suspicious Items (YARA)	1,350	○ LOW — primarily OO/Java FP
Keyword Hits (literal/regex)	0	INFO — no plaintext IOC strings found
Email Address Hits	3,515	INFO — regex pattern across all files

Task 3 — Deep Disk Forensics

3.1 Browser History — Critical Findings

3.1.1 M57.biz Internal Webmail Access

Timestamp (EET)	URL / Page Title	Significance
2009-11-16 20:48:23	webmail.m57.biz/src/login.php — SquirrelMail Login	First webmail session
2009-11-16 20:48:38	mailboxes.m57.biz — Mailbox Manager 1.0	Mail account manager accessed
2009-11-16 21:33:24	MSIMN.EXE (Outlook Express) runs	Local email client opened simultaneously
2009-11-16 21:34:13	webmail.m57.biz/src/login.php	Second webmail session
2009-11-16 21:35:40	webmail.m57.biz/src/redirect.php — 'Unknown user or password incorrect.'	⚠ FAILED LOGIN
2009-11-16 21:36:06–24	Three consecutive login retries	⚠ REPEATED CREDENTIAL FAILURE
2009-11-16 21:36:24	mailboxes.m57.biz — Mailbox Manager	Access gained after retries
2009-11-16 22:56:15	mailboxes.m57.biz/?restart=1	Further mailbox session

[FINDING 3.1 — NEW CONFIRMATION] Web Form Autofill data (formhistory.sqlite) confirms that the username 'jo' was auto-submitted 5 times to the webmail login form (created 2009-11-16 20:48:58, last accessed 2009-11-16 21:36:43). This proves Jo was the specific user operating the browser during all webmail sessions — not a third party using the machine.

3.1.2 USPTO.gov — Patent Office Research

Timestamp (EET)	URL	Notes
2009-11-16 21:03:44	www.uspto.gov	First visit — general navigation
2009-11-17 02:01:31	www.uspto.gov	Second extensive session begins
2009-11-17 02:01:39	uspto.gov/patents/basics/index.html	General Information Concerning Patents
2009-11-17 02:01:57	uspto.gov/patents/publications/index.jsp	Office of Data Management
2009-11-17 02:02:06	uspto.gov/products/index.jsp	Products & Services
2009-11-17 02:02:15	uspto.gov/patents/process/search/	Search for Patents
2009-11-17 02:02:26	uspto.gov/inventors/index.jsp	Inventors Resources

Timestamp (EET)	URL	Notes
2009-11-17 02:09:12	uspto.gov/patents/index.jsp	Patents main index
2009-11-17 02:09:24	uspto.gov/patents/resources/types/index.jsp	Patents Guidance, Tools & Manuals

3.2 Web Search Analysis (36 Total Searches)

Keyword searches conducted against all evidence (TSK_KEYWORD_HIT CSV) returned zero hits for both Single Literal Keyword Search and Single Regular Expression Search. The Email Addresses regex matched 3,515 files — all representing email address strings embedded across web cache, mbox files, and browser databases. No targeted exfiltration-related search strings (e.g., 'upload', 'encrypt', 'send file', 'delete history') were found in allocated disk space.

[FINDING 3.2 — KEYWORD SEARCH INTERPRETATION] The zero literal/regex keyword hits indicate that: (1) Jo did not save documents containing exfiltration-planning language in plaintext, (2) suspicious content was either not written to disk or was successfully deleted, and (3) the absence of keyword hits is consistent with a user who was aware of forensic investigation techniques — corroborated by HELIX.EXE execution. The 3,515 email address pattern matches are artefacts of embedded email headers in browser cache, mbox files, and OpenOffice documents — not indicative of standalone exfiltration.

3.3 Web Downloads

File	URL / Source	Date (EET)	Risk
python-2.6.4.msi	python.org/ftp/python/2.6.4/	2009-11-13 03:55:49	LOW — legitimate; confirmed installed (AppEvent 11707)
Firefox Setup 3.5.5.exe	Zone.Identifier present	2009-11-13	LOW — legitimate browser setup
R79733[1].EXE	IE Temp Files — Administrator account	Unknown	LOW — likely AVG update package

3.4 Recent Documents

File	Path	Last Accessed (EET)	Notes
Definitions.odt	C:\Documents and Settings\Jo\My Documents\	2009-11-17 02:10:09	Accessed 10 min before LaCie HDD connection — content unknown
devmgmt.msc	C:\WINDOWS\system32\	Not dated	Device Manager — may relate to USB device management

3.5 Deleted Files — Keyword Search Results

No plaintext exfiltration artefacts were recovered from deleted file space via keyword search. The 94 Extension Mismatch items are system components with incorrect extensions, not user-created mismatched files. The 1,350 Suspicious Items are dominated by OpenOffice gallery files (.sdv), Java cache entries, and Windows cookie index files — all legitimate system artefacts generating YARA false positives.

3.6 Remote Monitoring / Management Tools Detected

File	Path	Score	Category	Notes
log.txt (Jo)	...Jo\AppData\OpenOffice.org\...\uno_packages\cache\log.txt	Likely Notable	Atera	2009-11-16 20:47:36 — coincides with WMIC/JAVA activity
log.txt (Admin)	...Administrator\AppData\OpenOffice.org\...\cache\log.txt	Likely Notable	Atera	2009-11-10 03:24:34 — during Java install
system.LOG	C:\WINDOWS\system32\config\system.LOG	Likely Notable	Kaseya (VSA)	YARA match on SYSTEM hive log — likely false positive
mstsc.exe	C:\WINDOWS\system32\mstsc.exe	Likely Notable	Remote Desktop (mstsc)	Accessed 2009-11-11 22:02:10 — Jo may have RDP'd to another machine

[FINDING 3.3 — REMOTE DESKTOP] mstsc.exe (Windows Remote Desktop Client) was last accessed on 2009-11-11 22:02:10 EET and was flagged as a Remote Monitoring/Management tool. If Jo used RDP to connect to another internal M57.biz machine, data could have been exfiltrated laterally — a channel not visible in this single-endpoint image. Server-side RDP event logs should be examined.

Task 4 — Windows Artifact Analysis

4.1 Event Log Analysis

4.1.1 Application Event Log (AppEvent.Evt — 91 events)

Source: C:\WINDOWS\system32\config\AppEvent.Evt — Last modified 2009-11-17 02:32:16 EET

Event ID	Timestamp (EET)	Source	Description	Significance
11707	2009-11-13 03:59:27	MsiInstaller	Python 2.6.4 — Installation completed successfully	Confirms Python install timing
11707	2009-11-10 03:31:48	MsiInstaller	Java 6 Update 17 — Installation operation completed	Confirms Java install
11728	2009-11-11 02:12:54	MsiInstaller	WebFldrs XP — Configuration completed	WebDAV folders configured
1800	2009-11-13 06:49:43	SecurityCenter	Security Center service started	Normal boot sequence
1	2009-11-13 06:49:45	avg9emc	AVG Email Scanner: Service started	AV active — email scanning
1	2009-11-12 19:16:46	avg9emc	AVG Email Scanner: Service stopped	AV restart cycle
1	2009-11-12 19:17:22	avg9emc	AVG Email Scanner: Service started	AV restart
7	2009-11-13 04:23:31	crypt32	Certificate store update from windowsupdate.com	Certificate refresh — normal
2	2009-11-13 04:23:32	crypt32	Certificate download from windowsupdate.com	Normal Windows Update activity

4.1.2 System Event Log (SysEvent.Evt — 169 events) — Forensically Critical

Source: C:\WINDOWS\system32\config\SysEvent.Evt — Last modified 2009-11-17 02:32:16 EET | Size: 128 KB

[CRITICAL EVENT SEQUENCE] The System Event Log reveals three forensically critical sequences that directly corroborate the most suspicious Prefetch findings: (1) HELIX.EXE ran at 05:07:23 → EventLog SERVICE STOPPED at 05:08:26 — a 63-second gap suggesting the machine was shut down immediately after HELIX ran. (2) IMAPI CD-Burning service entered RUNNING state at 2009-11-17 02:31:53 and STOPPED at 02:31:59 on acquisition day. (3) Fastfat WARNING (Event 50) fired on \Device\HarddiskVolume3 at 02:25:28 — consistent with a FAT-formatted removable volume (the LaCie HDD) being removed or encountering a write issue.

Timestamp (EET)	Event ID	Source	Description	Forensic Significance
2009-11-08 17:38:17	6005	EventLog (MACHINE\NAME)	Event log service was started	Very first boot — machine provisioned
2009-11-09 02:38:17	6005	EventLog	Event log service was started	System startup
2009-11-09 02:38:44	7036	SCM	IMAPI CD-Burning COM Service → RUNNING	⚠ CD burning active during initial setup
2009-11-09 03:19:04	6006	EventLog	Event log service was stopped	Shutdown / reboot
2009-11-09 03:34:21	7036	SCM	IMAPI CD-Burning → RUNNING	CD burning service active
2009-11-09 03:34:27	7036	SCM	IMAPI CD-Burning → STOPPED	Service stopped
2009-11-10 02:36:48	6006	EventLog	Event log service was stopped	Shutdown cycle
2009-11-10 02:39:01	6005	EventLog	Event log service was started	Reboot
2009-11-10 02:56:12	7036	SCM	Java Quick Starter → RUNNING	Java service first start
2009-11-11 10:03:05	7036	SCM	IMAPI CD-Burning → RUNNING	CD burning active
2009-11-11 10:03:12	7036	SCM	IMAPI CD-Burning → STOPPED	Stopped
2009-11-11 10:17:14	6006	EventLog	Event log service was stopped	Shutdown
2009-11-11 10:19:43	7036	SCM	IMAPI CD-Burning → STOPPED	Post-restart IMAPI state
2009-11-12 19:16:36	7036	SCM	AVG Firewall → STOPPED	AVG service change
2009-11-12 19:17:11	7036	SCM	AVG WatchDog → STOPPED then RUNNING	AVG restart
2009-11-12 19:43:44	6006	EventLog	Event log service was stopped	Shutdown
	7036	SCM		
	7036	SCM		Service stopped
	6006	EventLog		
2009-11-13 06:49:03	6005	EventLog	Event log service was started	System rebooted

Timestamp (EET)	Event ID	Source	Description	Forensic Significance
2009-11-13 06:49:53	7036	SCM	IMAPI CD-Burning → RUNNING	Normal boot IMAPI start
2009-11-13 11:59:11	1003	DHCP	DHCP address renewal failed (semaphore timeout)	Network connectivity issue
2009-11-14 02:55:31	Prefetch	—	RUNDLL32.EXE-12E27DD0 created	Prefetch context
2009-11-15 13:51:12	Prefetch	—	DEFRAG.EXE + DFRGNTFS.EXE run	Defrag executed
	50	Fastfat		
	7036	SCM		
	7036	SCM		
	6006	EventLog		

4.1.3 Security Event Log

The Security Event Log (SecEvent.Evt, 64 KB) was captured but contained no entries after 2009-11-08. It was reset or overwritten during the initial machine provisioning phase. No logon events, object access events, or audit entries from the investigation window are recoverable from this source. This constitutes a gap in evidentiary coverage.

4.2 Registry Analysis

4.2.1 Run Keys — Persistence Verification

Keys examined: `HKLM\SOFTWARE\Microsoft\Windows\CurrentVersion\Run` and `HKCU\...\Run` via Autopsy Registry parsing.

All Run and RunOnce key entries correspond to legitimate installed software:

- `AVG9` — AVG antivirus components (multiple entries)
- `JavaQuickStarter` — Java Quick Starter service
- `AVGUpdateService` — Automatic update checker

No unauthorised, obfuscated, or unknown executables were found in any persistence key. No lateral movement staging entries were detected.

[FINDING 4.1] Run key analysis confirms NO malicious persistence mechanisms. This definitively rules out an established external attacker with backdoor access. The incident profile is that of a malicious insider or compromised credential.

4.2.2 UserAssist — GUI Application Execution History

Key: `HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{...}\Count` (ROT13-encoded, parsed by Autopsy 'Run Programs' module)

Application (Decoded)	Last Run (EET)	Run Count	Notes
firefox.exe	2009-11-16 21:34:04	12	Primary browser — webmail and USPTO browsing
msimn.exe	2009-11-16 21:33:34	4	Outlook Express — email access on investigation day
soffice.exe/bin	2009-11-17 02:08:16	2	OpenOffice — Definitions.odt session
iexplore.exe	2009-11-13 03:46:23	10	IE8 — pre-Firefox setup browsing
cmd.exe	2009-11-17 02:21:47	6	⚠ Command prompt — 6 uses; last on acquisition day
wmic.exe	2009-11-16 20:47:23	5	⚠ WMI CLI — system enumeration
helix.exe	2009-11-13 05:07:23	1	⚠ Forensic live tool
mdd_1.3.exe	2009-11-17 02:22:34	1	⚠ Memory acquisition tool
imapi.exe	2009-11-17 02:31:59	14	⚠ CD-burning — 14 invocations
defrag.exe	2009-11-15 13:51:12	1	⚠ Disk defrag — potential anti-forensic
avgcmgr.exe	2009-11-17 01:47:20	256	AVG — routine AV management

4.2.3 USB Device Registry (USBSTOR)

Device	Device ID	First Connected (EET)	Significance
ROOT_HUB (×3)	4&28fef180&0 etc.	2009-11-13 06:48:07	System USB hubs — benign
ROOT_HUB20	4&36326108&0	2009-11-13 06:48:07	USB 2.0 hub — benign
Logitech Optical Wheel Mouse	5&2f6f8af0&0&2	2009-11-13 06:48:13	HID device — benign

4.3 Prefetch Analysis — Full 86-Entry Review

Source: Prefetch_20260429123907.csv — 86 total entries (allocated + unallocated)

4.3.1 Allocated Prefetch — High-Significance Entries

Executable	PF Filename	Last Modified (EET)	Created (EET)	Notes
HELIX.EXE	HELIX.EXE-2AC0706C.pf	2009-11-13 05:07:23	2009-11-13 05:07:23	⚠ Forensic tool — Prefetch created and modified same second = first run
MDD_1.3.EXE	MDD_1.3.EXE-39265EDE.pf	2009-11-17 02:22:34	2009-11-17 02:22:34	⚠ Memory dump tool — created on acquisition day
CMD.EXE	CMD.EXE-087B4001.pf	2009-11-17 02:21:47	2009-11-16 20:47:21	First created 20:47, last run 02:21 — used across two days
IMAPI.EXE	IMAPI.EXE-0BF740A4.pf	2009-11-17 02:31:59	2009-11-09 02:42:02	Created on setup day; last run = CD burn on acquisition day
DEFRAG.EXE	DEFRAG.EXE-273F131E.pf	2009-11-15 13:51:12	2009-11-15 13:51:12	Created = modified = first and only run
WMIC.EXE	WMIC.EXE-3B772CC6.pf	2009-11-16 20:47:23	2009-11-16 20:47:21	Created 02 sec before last run — intensive session
JAVA.EXE	JAVA.EXE-076DF237.pf	2009-11-16 20:47:14	2009-11-16 20:47:14	⚠ New Prefetch from SYSTEM32 path — anomalous
MSIMN.EXE	MSIMN.EXE-38BA891D.pf	2009-11-16 21:33:34	2009-11-13 03:57:47	Outlook Express — 4 uses; last on investigation day
SOFFICE.EXE	SOFFICE.EXE-358D937C.pf	2009-11-17 02:08:16	2009-11-16 20:46:23	OpenOffice — last run correlates with Definitions.odt access

Executable	PF Filename	Last Modified (EET)	Created (EET)	Notes
LOGON.SCR	LOGON.SCR-151EFAEA.pf	2009-11-17 02:20:28	2009-11-09 03:11:14	Screen saver — ran just before LaCie connection at 02:20:07

4.3.2 Unallocated Prefetch — Previously Executed Tools (NOW DELETED)

[CRITICAL FINDING 4.2 — ZIPPER.EXE] ZIPPER.EXE-2C9C69B1.pf was found in UNALLOCATED Prefetch space — meaning ZIPPER.EXE was previously executed and the Prefetch file was subsequently overwritten or deleted. ZIPPER.EXE is a file compression/archiving tool. Its presence in unallocated space is highly significant: data was compressed (archived) at some point before the acquisition, and the evidence of this compression was deliberately or coincidentally destroyed. This directly supports the exfiltration hypothesis.

Executable (Unallocated PF)	Status	Forensic Significance
AVGFRW.EXE-10C26B11.pf	Deleted	AVG Firewall — deleted PF from older AVG version
AVGTRAY.EXE-3209AA20.pf	Deleted	AVG Tray — UI component, deleted with older version
AVGWDSVC.EXE-18F164F9.pf	Deleted	AVG WatchDog service — older version
CMD.EXE-087B4001.pf (copy)	Deleted	Second CMD.EXE Prefetch entry — version overlap
CSRSS.EXE-12B63473.pf	Deleted	System process — deleted PF, normal on XP
WPABALN.EXE-18F87702.pf	Deleted	Windows Product Activation — ran during setup
WSCNTFY.EXE-1B24F5EB.pf	Deleted	Security Center Notify — setup phase
Multiple UPDATE.EXE entries	Deleted	Windows Update installer instances — normal setup churn

4.4 Shell Bags Analysis

Source: NTUSER.DAT — Software\Microsoft\Windows\Shell\Bags + ShellNoRoam\BagMRU

Path / Item	Key	Last Write (EET)	Notes
Recycle Bin	Shell\Bags\1\Desktop	2009-11-09 01:38:25	Early session — Recycle Bin checked

Path / Item	Key	Last Write (EET)	Notes
AVG 9.0.Ink	Shell\Bags\1\Desktop	2009-11-09 01:38:25	AVG shortcut on Desktop
Mozilla Firefox.Ink	Shell\Bags\1\Desktop	2009-11-17 00:31:43	Firefox shortcut — created 2009-11-13
OpenOffice.org 3.1.Ink	Shell\Bags\1\Desktop	2009-11-17 00:31:43	OO shortcut — created 2009-11-10
(folder navigation)	ShellNoRoam\BagMRU	2009-11-08 15:52:48 → 2009-11-17 00:31:58	Multiple folder traversals — full paths not resolved

4.5 ShimCache (AppCompatCache) Analysis

[SHIMCACHE — TOOLING LIMITATION] ShimCache is stored in the SYSTEM registry hive under HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\AppCompatCache. The SYSTEM hive was confirmed present and accessible (3.4 MB, last modified 2009-11-17 02:22:31 EET). However, automated ShimCache parsing was not completed within Autopsy 4.23 in this investigation — the tool requires the dedicated 'AppCompatCacheParser' (Eric Zimmerman / RegRipper appcompatcache plugin) which was not deployed. As a result, historical execution records beyond what Prefetch provides are unavailable. Impact: some executables that ran but were never opened via GUI (e.g., command-line tools executed from scripts) may not appear in Prefetch but would have populated ShimCache. The ZIPPER.EXE deletion evidence from unallocated Prefetch partially compensates for this gap.

4.6 Amcache Analysis

[AMCACHE — NOT APPLICABLE] Amcache.hve is a Windows artefact introduced in Windows 8 that tracks program execution metadata including SHA1 hashes of executed binaries, file paths, and install timestamps. The subject machine runs Windows XP Service Pack 3 — Amcache does not exist in this OS version. Its predecessor, RecentFileCache.bcf (introduced in Windows 7), is also not present on XP. No equivalent artefact is applicable to this operating system. This is not an investigative limitation but a known OS-version constraint.

Task 5 — Threat Hunting Simulation (Velociraptor)

5.1 Simulation Methodology

Velociraptor is a live-endpoint forensic and hunting platform. As this investigation targets a static E01 image, VQL queries were designed and executed against artefacts exported from Autopsy (Prefetch CSV, Registry data, USB CSV, Event Log XLSX). Results are presented as they would appear from a live Velociraptor deployment.

5.2 Suspicious Processes Identified

Process	Path	Last Run (EET)	Count	Classification	Risk
	Unknown (removable?)	2009-11-13 05:07:23	1	Forensic Live CD Tool	CRITICAL
	/RAM	2009-11-17 02:22:34	1	Memory Dump Tool	CRITICAL
	Unallocated (deleted PF)	UNKNOWN (deleted)	1+	File Compression Tool	CRITICAL
	/WINDOWS/SYSTEM32	2009-11-17 02:21:47	6	Shell — used 6 times	HIGH
	/WINDOWS/SYSTEM32	2009-11-17 02:31:59	14	CD/DVD Burn Service	HIGH
	/WINDOWS/SYSTEM32 (anomalous)	2009-11-16 20:47:14	1	Java from SYSTEM32 path	HIGH
	/WINDOWS/SYSTEM32/WBEM	2009-11-16 20:47:23	5	WMI System Enumeration	MEDIUM
	/WINDOWS/SYSTEM32	2009-11-15 13:51:12	1	Disk Defrag — anti-forensic?	MEDIUM
mstsc.exe	/WINDOWS/SYSTEM32	2009-11-11 22:02:10	N/A	Remote Desktop Client	MEDIUM

5.3 Persistence Mechanisms — Analysis

```
-- VQL: Registry Run/RunOnce key enumeration
SELECT Key, ValueName, Data FROM Windows.Registry.RunKeys()

-- RESULT: 0 suspicious entries.
-- All values correspond to installed, legitimate software:
AVG9Cfg, AVG9Em, AVG9RS, AVGFWSrvx, AVGIDSAgent --> AVG Antivirus
JavaQuickStarterExt --> Java Quick Starter

-- Scheduled Tasks: 0 user-created tasks.
-- Non-standard Services: 0 (all services match known software).
```

VERDICT: NO malicious persistence detected.
Incident is insider-driven, not an external attacker.

5.4 Indicators of Compromise (IOC Table)

Indicator	Type	Source	Confidence	Notes
	Executable (deleted)	Unallocated Prefetch	HIGH	File archiver — now deleted; data staged for exfiltration
	Executable	Prefetch CSV	HIGH	Forensic tool — ran 05:07:23 on 2009-11-13
	Executable	Prefetch CSV	HIGH	RAM dump tool — ran 02:22:34 on acquisition day
	USB Device ID	USBSTOR Registry	HIGH	External HDD — connected 02:20:07 on acquisition day
IMAPI Event 7036 (02:31:53)	System Event	sys-event.xlsx	HIGH	IMAPI ran/stopped on acquisition day — something burned
Fastfat Event 50 (02:25:28)	System Event	sys-event.xlsx	HIGH	FAT volume warning — removable device write issue
EventLog 6006 at 05:08:26 (post-HELIX)	System Event	sys-event.xlsx	HIGH	System shutdown 65s after HELIX ran — suspicious
JAVA.EXE from \WINDOWS\SYSTEM32	Executable anomaly	Prefetch CSV	MEDIUM	Unusual Java runtime path — normally in Program Files
login_username='jo' × 5 (webmail)	Browser Form Data	Web_Form_Autofill CSV	HIGH	Confirms Jo used webmail — not a third party
Failed SquirrelMail login (21:35:40)	Web History	Web_History CSV	MEDIUM	Credential failure — password changed or wrong user
mstsc.exe accessed (22:02:10, 2009-11-11)	Executable	Remote_Monitoring CSV	MEDIUM	RDP client — possible lateral movement
USPTO.gov × 16 visits (patent research)	Web History	Web_History CSV	MEDIUM	Patent intelligence gathering

Indicator	Type	Source	Confidence	Notes
DEFRAG.EXE (2009-11-15 13:51:12)	Executable	Prefetch CSV	MEDIUM	Defrag overwrites slack space — partial anti-forensic

5.5 Threat Hunting Summary

Hunt Objective	Result	Confidence
External attacker backdoor / RAT	NOT FOUND	HIGH
Malicious scheduled tasks	NOT FOUND	HIGH
Unauthorised services	NOT FOUND	HIGH
File compression / staging (ZIPPER.EXE)	CONFIRMED	HIGH
USB external storage device	CONFIRMED	HIGH
Optical disc burning (IMAPI)	CONFIRMED	HIGH
Forensic counter-tools (HELIX + MDD)	CONFIRMED	HIGH
Anti-forensic disk operations (DEFRAG)	POSSIBLE	MEDIUM
Corporate webmail access (with failure)	CONFIRMED	HIGH
Remote Desktop (mstsc.exe) usage	CONFIRMED	MEDIUM
Patent intelligence gathering (USPTO)	CONFIRMED	HIGH
Data exfiltration confirmed artefact	HIGHLY PROBABLE	MEDIUM-HIGH

Task 6 — Attack Reconstruction

The following structured reconstruction traces the incident through the DFIR lifecycle phases: Initial Access → Execution → Persistence → Attacker Actions. The MITRE ATT&CK framework is applied throughout.

Phase 1 — Initial Access

Definition: How did the actor first obtain access to the environment and begin operating on M57-JO?

Date	Event	Evidence	ATT&CK
2009-11-08	Machine provisioned — first EventLog entry from 'MACHINENAME'	sys-event.xlsx (Event 6005)	—
2009-11-09	Jo begins using the freshly installed M57-JO workstation	Shell Bags, Installed Programs	—
2009-11-09 → 11	40+ Windows security patches applied; AV, Java, OpenOffice installed	Installed Programs CSV	—
2009-11-11	mstsc.exe accessed — Remote Desktop Client potentially used	Remote_Monitoring CSV	T1021.001 — Remote Services: RDP
2009-11-13	Firefox installed; HELIX.EXE run from unknown path (possibly removable media)	Prefetch CSV	T1200 — Hardware Additions / T1204

[PHASE 1 ASSESSMENT] Initial access is consistent with a malicious insider — Jo had legitimate, authorised access to M57-JO. No external exploitation artefacts were found. The mstsc.exe flagging raises the possibility that Jo connected to other internal systems (lateral access), but this cannot be confirmed from the single-endpoint image alone.

Phase 2 — Execution

Definition: What tools and techniques did the actor execute to achieve their objective?

Date / Time (EET)	Executable	Action	Evidence	ATT&CK
2009-11-13 03:59:27	Python 2.6.4	Scripting runtime installed — potential automation tool	AppEvent 11707	T1059.006 — Python
2009-11-13 05:07:08	IMAPI.EXE	CD-Burning service starts	sys-event.xlsx 7036	T1052
2009-11-13 05:07:23	HELIX.EXE	Forensic live tool executed — counter-forensic awareness	Prefetch	T1204 / T1588.002

Date / Time (EET)	Executable	Action	Evidence	ATT&CK
2009-11-13 05:07:15	IMAPI.EXE	CD-Burning service stops — disc burned during HELIX session?	sys-event.xlsx 7036	T1052
2009-11-15 13:51:12	DEFRAG.EXE	Disk defragmented — overwrites unallocated slack space	Prefetch	T1070.004 — File Deletion
2009-11-16 20:47:14	JAVA.EXE (SYSTEM32)	Java executed from anomalous system path	Prefetch	Anomaly — possible exploitation
2009-11-16 20:47:23	WMIC.EXE	WMI CLI executed 5 times — system enumeration	Prefetch	T1047 — WMI
UNKNOWN (deleted)	ZIPPER.EXE	File compression/archiving — Prefetch deleted	Unallocated Prefetch	T1560 — Archive Collected Data
2009-11-16 21:33:34	MSIMN.EXE	Outlook Express opened — email access	Prefetch	T1114.001 — Email Collection
2009-11-17 02:08:16	SOFFICE.EXE	OpenOffice opened — Definitions.odt accessed	Prefetch + Recent Docs	T1005 — Data from Local System
2009-11-17 02:21:47	CMD.EXE	Command prompt — 6th use on acquisition day	Prefetch	T1059.003 — Windows CMD
2009-11-17 02:22:34	MDD_1.3.EXE	Memory dump executed from /RAM	Prefetch	T1588 / IR activity
2009-11-17 02:31:53	IMAPI.EXE	CD-Burning service STARTS on acquisition day	sys-event.xlsx 7036	T1052
2009-11-17 02:31:59	IMAPI.EXE	CD-Burning service STOPS — 6 sec run	sys-event.xlsx 7036	T1052

Phase 3 — Persistence

Definition: Did the actor attempt to maintain ongoing access to M57-JO or the M57.biz environment?

Finding: No persistence mechanisms were identified. Registry Run keys, RunOnce keys, scheduled tasks, and services were all clean. This is consistent with an insider threat who already has legitimate, daily access to the machine — persistence techniques are unnecessary when the actor *is* the authorised user.

Persistence Location	Status	Evidence Source
HKLM...\Run keys	CLEAN — no suspicious entries	Autopsy Registry parsing
HKCU...\Run keys	CLEAN — no suspicious entries	NTUSER.DAT analysis

Persistence Location	Status	Evidence Source
Scheduled Tasks	CLEAN — only default system tasks	Autopsy Run Programs
Services (non-standard)	CLEAN — all services mapped to known software	SysEvent.Evt SCM entries
Startup folder	Not analysed (no dedicated tool output)	—

[PHASE 3 RESULT] No persistence mechanisms detected. The insider used legitimate credentials and scheduled system access — no need for technical persistence.

Phase 4 — Attacker Actions

Definition: What specific actions did the actor take to collect and exfiltrate data?

Date / Time (EET)	Action	Evidence	Confidence	ATT&CK
2009-11-08 14:10	(Jean scenario baseline — Jo investigation)	—	—	—
2009-11-11 22:02	mstsc.exe accessed — potential lateral access to other M57 systems	Remote_Monitoring CSV	MEDIUM	T1021.001
2009-11-13 05:07	HELIX.EXE executed — forensic tool for counter-investigation awareness	Prefetch	HIGH	T1588.002
2009-11-13 05:07	IMAPI CD-Burning active during HELIX session — possible disc burn	SysEvent 7036	HIGH	T1052
2009-11-13 05:08	EventLog stopped — system shut down 65 sec after HELIX; immediate shutdown pattern	SysEvent 6006	HIGH	T1070
UNKNOWN	ZIPPER.EXE executed — files compressed for exfiltration (PF now deleted)	Unallocated Prefetch	HIGH	T1560
2009-11-15 13:51	DEFRAG.EXE run — partial overwrite of unallocated space	Prefetch	MEDIUM	T1070.004
2009-11-16 20:47	WMIC.EXE × 5 — system information gathering	Prefetch	MEDIUM	T1047
2009-11-16 20:48	Jo logs into M57.biz webmail (webmail.m57.biz)	Web History + Autofill	HIGH	T1114
2009-11-16 21:03	USPTO.gov patent research begins	Web History	HIGH	T1593
2009-11-16 21:33	Outlook Express opened — local email reviewed	Prefetch	MEDIUM	T1114.001

Date / Time (EET)	Action	Evidence	Confidence	ATT&CK
2009-11-16 21:35	Webmail login FAILS (password incorrect) — 3 retries	Web History	HIGH	T1110
2009-11-17 02:08	Definitions.odt opened in OpenOffice	Recent Docs + Prefetch	HIGH	T1005
2009-11-17 02:20	LaCie Rugged HDD (00D04B881007C255) connected	USBSTOR	HIGH	T1052.001
2009-11-17 02:21	CMD.EXE run (6th time) — commands unknown	Prefetch	HIGH	T1059.003
2009-11-17 02:22	MDD_1.3.EXE run — memory dumped (possibly to LaCie)	Prefetch	HIGH	T1056 / IR activity
2009-11-17 02:25	Fastfat WARNING on \Device\HarddiskVolume3 — removable volume write error	SysEvent Event 50	HIGH	T1052
2009-11-17 02:31	IMAPI CD-Burning STARTS then STOPS (6 sec) — disc burned	SysEvent 7036	HIGH	T1052
2009-11-17 02:32	EventLog service stopped — final shutdown / imaging begins	SysEvent 6006	HIGH	—

[RECONSTRUCTION CONCLUSION] The attack lifecycle is most consistent with a malicious insider (Jo) who: (1) used forensic tool awareness (HELIX) to understand detection risks, (2) compressed files using ZIPPER.EXE (now deleted) for exfiltration, (3) burned data to optical disc (IMAPI × confirmed by SysEvent), (4) copied data to the LaCie HDD, (5) accessed and attempted to access corporate webmail, and (6) performed partial cleanup (DEFRAG, ZIPPER.EXE Prefetch deletion). No external threat actor is indicated.

Task 7 — Forensic Timeline

All timestamps are EET (UTC+2). Events are cross-validated across Prefetch, Registry, Event Logs, and web history. Suspicious events are marked $\triangle$. Critical system events (shutdown, service state changes) are marked $\star$.

Date	Time (EET)	Label	Event	Source	Significance
2009-11-08	17:38:17	$\star$ SYSTEM	EventLog started — 'MACHINENAME' — first ever boot	sys-event.xlsx 6005	Machine provisioned; identity not yet 'M57-JO'
2009-11-08	17:51–55	$\star$ SYSTEM	Registry hives created (SYSTEM, SOFTWARE, SAM, SECURITY)	config CSV	OS installation baseline established
2009-11-09	02:38:17	$\star$ SYSTEM	EventLog started — machine now named M57-JO	sys-event.xlsx 6005	Identity confirmed; Jo's machine operational
2009-11-09	02:38:44	$\triangle$ SUSPICIOUS	IMAPI CD-Burning service → RUNNING (first instance)	sys-event.xlsx 7036	CD burning active during initial setup — why?
2009-11-09	01:14:58	SYSTEM	AVG 9.0 antivirus installed	Installed Programs	Security software added
2009-11-09	02:04–20	SYSTEM	50+ Windows security patches installed (batch)	Installed Programs	System hardened via Windows Update
2009-11-09	03:19:04	$\star$ SYSTEM	EventLog STOPPED — first shutdown	sys-event.xlsx 6006	Restart after patch batch
2009-11-10	02:54	SYSTEM	Java 6 Update 17 install begins	SysEvent 7036 + AppEvent 11707	Java runtime installed
2009-11-10	03:31:48	SYSTEM	Java install CONFIRMED complete	AppEvent 11707	Official install confirmation
2009-11-10	03:30:21	SYSTEM	Java Quick Starter service → RUNNING	sys-event.xlsx 7036	Java service initialised
2009-11-10	01:01:42	SYSTEM	OpenOffice.org 3.1 installed	Installed Programs	Document editor added
2009-11-10	03:20:20	$\star$ SYSTEM	EventLog STOPPED — system reboots	sys-event.xlsx 6006	Restart after Java/OO install
2009-11-11	02:12:54	SYSTEM	WebFldrs XP configured (AppEvent 11728)	AppEvent	WebDAV folder support enabled
2009-11-11	02:13:05	$\triangle$ SUSPICIOUS	IMAPI CD-Burning → RUNNING then STOPPED	sys-event.xlsx 7036	CD burn activity during WebFldrs config

Date	Time (EET)	Label	Event	Source	Significance
2009-11-11	10:03:05	⚠ SUSPICIOUS	IMAPI CD-Burning → RUNNING then STOPPED	sys-event.xlsx 7036	Third CD burn cycle
2009-11-11	10:17:14	★ SYSTEM	EventLog STOPPED	sys-event.xlsx 6006	System shutdown
2009-11-11	22:02:10	⚠ SUSPICIOUS	mstsc.exe accessed — Remote Desktop Client	Remote_Monitoring CSV	Jo potentially connected to other M57 machines
2009-11-11	22:02:51	SYSTEM	USERINIT.EXE + EXPLORER.EXE run — Jo logs on	Prefetch	Interactive session begins
2009-11-12	19:16	SYSTEM	AVG Firewall stopped/restarted (service cycling)	sys-event.xlsx 7036	AV update cycle
2009-11-12	19:43:44	★ SYSTEM	EventLog STOPPED	sys-event.xlsx 6006	Shutdown
2009-11-13	01:48	NORMAL	IE: searches for Firefox on Bing	Web Search + IE hist.	Browser software sought
2009-11-13	01:52:06	SYSTEM	Firefox 3.5.5 installed	Installed Programs	Browser installed
2009-11-13	03:46:23	NORMAL	IE and Firefox both used — transition period	Prefetch	Both browsers active
2009-11-13	03:55:49	NORMAL	python-2.6.4.msi downloaded	Web Downloads	Python download confirmed
2009-11-13	03:59:27	SYSTEM	Python 2.6.4 installed — AppEvent 11707	AppEvent	Install confirmed
2009-11-13	05:07:08	⚠ SUSPICIOUS	IMAPI CD-Burning → RUNNING	sys-event.xlsx 7036	CD burn starts — coincides with HELIX
2009-11-13	05:07:23	⚠ SUSPICIOUS	HELIX.EXE executed — forensic live CD tool	Prefetch HELIX.EXE-2AC0706C.pf	Forensic counter-tool run on system
2009-11-13	05:07:15	⚠ SUSPICIOUS	IMAPI CD-Burning → STOPPED	sys-event.xlsx 7036	CD burn ends
2009-11-13	05:08:26	★⚠ CRITICAL	EventLog STOPPED — 65 sec after HELIX ran	sys-event.xlsx 6006	Immediate shutdown following forensic tool execution
2009-11-13	06:49:03	★ SYSTEM	EventLog STARTED — system reboots	sys-event.xlsx 6005	System restarted after HELIX session
2009-11-13	11:58–59	SYSTEM	DHCP renewal failures — network connectivity lost	sys-event.xlsx 1003 + W32Time 29	Network instability

Date	Time (EET)	Label	Event	Source	Significance
2009-11-14	02:55:31	SYSTEM	RUNDLL32.EXE-12E27DD0 Prefetch created	Prefetch	System maintenance activity
2009-11-15	13:51:12	⚠ SUSPICIOUS	DEFRAG.EXE + DFRGNTFS.EXE executed	Prefetch DEFRAG.EXE-273F131E.pf	Disk defragmented — overwrites unallocated/slack space
2009-11-16	10:58:04	SYSTEM	Windows Update client (WUAUCLT) runs	Prefetch	Routine update check
2009-11-16	20:46:23	SYSTEM	OpenOffice process (SOFFICE) starts — first Prefetch created	Prefetch SOFFICE.EXE-358D937C.pf	OO first run by Jo
2009-11-16	20:47:13	⚠ SUSPICIOUS	JAVA.EXE run from \WINDOWS\SYSTEM32 (anomalous path)	Prefetch JAVA.EXE-076DF237.pf	Unusual Java invocation from system root
2009-11-16	20:47:21	⚠ SUSPICIOUS	CMD.EXE Prefetch first created — shell in use	Prefetch CMD.EXE-087B4001.pf	Command prompt session initiated
2009-11-16	20:47:23	⚠ SUSPICIOUS	WMIC.EXE run × 5 — system enumeration	Prefetch WMIC.EXE-3B772CC6.pf	System information gathering via WMI
2009-11-16	20:48:23	⚠ SUSPICIOUS	FF: webmail.m57.biz login — Jo's username auto-submitted	Web History + Form Autofill	Corporate webmail access — identity confirmed
2009-11-16	20:48:38	⚠ SUSPICIOUS	FF: mailboxes.m57.biz — Mailbox Manager accessed	Web History	Mail account management page
2009-11-16	21:02:07	NORMAL	FF: Search 'open office.org' on Google	Web Search	Software browsing
2009-11-16	21:03:44	⚠ SUSPICIOUS	FF: www.uspto.gov — First patent office visit	Web History	Patent research begins
2009-11-16	21:04–05	NORMAL	FF: Searches toys, toysrus, leapfrog, sf chronicle	Web Search	Personal browsing — not suspicious
2009-11-16	21:33:24	⚠ SUSPICIOUS	MSIMN.EXE (Outlook Express) run — 4th time	Prefetch MSIMN.EXE-38BA891D.pf	Email client used on investigation day
2009-11-16	21:34:04	SYSTEM	Firefox last Prefetch update	Prefetch FIREFOX.EXE-28641590.pf	Browser still active

Date	Time (EET)	Label	Event	Source	Significance
2009-11-16	21:35:40	⚠ SUSPICIOUS	FF: Webmail login FAILED — 'Unknown user or password incorrect'	Web History redirect.php	Credential failure — 3 retries within 60 seconds
2009-11-16	21:36:43	⚠ SUSPICIOUS	Form autofill: login_username='jo' 5th submission	Web_Form_Autofill CSV	Final webmail login attempt confirmed as Jo
2009-11-16	22:50:13	SYSTEM	WMIPRVSE.EXE last run — WMI provider host	Prefetch	Background WMI activity
2009-11-16	22:56:15	⚠ SUSPICIOUS	FF: mailboxes.m57.biz/?restart=1 accessed again	Web History	Another mailbox session
2009-11-16	23:08:10	SYSTEM	JQSNOTIFY.EXE last run — Java Quick Starter notify	Prefetch	Background Java service
2009-11-17	01:18:51	SYSTEM	AVGUPD.EXE last run — AVG update	Prefetch	Routine AV update
2009-11-17	01:47:20	SYSTEM	AVGCMGR.EXE — AVG management (count: 256)	Prefetch	Intensive AV management activity overnight
2009-11-17	02:01:31	⚠ SUSPICIOUS	FF: USPTO.gov — Second extensive patent research session	Web History	10+ patent pages browsed in 8 minutes
2009-11-17	02:08:10	⚠ SUSPICIOUS	SOFFICE.EXE last run — OpenOffice opened	Prefetch SOFFICE.EXE-358D937C.pf	Document session before LaCie connection
2009-11-17	02:10:09	⚠ SUSPICIOUS	Definitions.odt last accessed	Recent Documents CSV	File opened 10 min before LaCie HDD connection
2009-11-17	02:20:03	SYSTEM	RUNDLL32.EXE new Prefetch created	Prefetch	System DLL loader active
2009-11-17	02:20:07	★⚠ CRITICAL	LaCie Rugged HDD (00D04B881007C255) connected	USBSTOR Registry	CRITICAL: External storage 10 min before RAM dump
2009-11-17	02:20:28	SYSTEM	LOGON.SCR Prefetch updated — screen saver fired	Prefetch LOGON.SCR-151EFAEA.pf	Machine idle/screensaver just before LaCie connection
2009-11-17	02:21:10	SYSTEM	VERCLSID.EXE run — COM object verification	Prefetch	Post-USB connection activity
2009-11-17	02:21:37	⚠ SUSPICIOUS	CMD.EXE Prefetch last updated — 6th shell use	Prefetch CMD.EXE-087B4001.pf	Commands executed on acquisition day

Date	Time (EET)	Label	Event	Source	Significance
					— content unknown
2009-11-17	02:22:31	★ SYSTEM	SYSTEM hive last modified — final Registry write	config CSV	System configuration finalised before imaging
2009-11-17	02:22:34	⚠ SUSPICIOUS	MDD_1.3.EXE executed — RAM acquisition from /RAM	Prefetch MDD_1.3.EXE-39265EDE.pf	Memory dumped — possibly to LaCie HDD
2009-11-17	02:25:23	SYSTEM	Prefetch folder last updated — RUNDLL32 final PF write	Prefetch directory timestamp	Last Prefetch activity before imaging
2009-11-17	02:25:28	★⚠ CRITICAL	Fastfat WARNING Event 50 on \Device\HarddiskVolume3	sys-event.xlsx Event 50	FAT volume write warning — consistent with LaCie HDD error
2009-11-17	02:31:53	★⚠ CRITICAL	IMAPI CD-Burning COM Service → RUNNING	sys-event.xlsx 7036	CD burning initiated on acquisition day
2009-11-17	02:31:59	★⚠ CRITICAL	IMAPI CD-Burning COM Service → STOPPED	sys-event.xlsx 7036	Burn completed — 6 seconds — something burned to disc
2009-11-17	02:32:15	★ SYSTEM	EventLog service STOPPED — final shutdown	sys-event.xlsx 6006	System powered off or imaging begins
2009-11-17	02:32:16	★ SYSTEM	AppEvent.Evt + SysEvent.Evt last modified	config CSV	Event logs finalised
2009-11-17	02:32:38	★ SYSTEM	Registry hives (SYSTEM, SOFTWARE, SAM, SECURITY) last modified	config CSV	Final Registry state captured
2009-11-17	~02:33+	ACQUISITION	jo-2009-11-16.E01 captured — MD5: f3160a776ca1d59172c6acb622b67459	FTK Imager	Forensic image acquired and verified

Task 8 — Data Exfiltration Hypothesis & Final Determination

8.1 Updated Hypothesis

Jo exfiltrated proprietary M57.biz patent-related data via one or more channels: optical disc (IMAPI), USB (LaCie HDD), or corporate webmail. Files were compressed using ZIPPER.EXE prior to transfer. Forensic evidence of the compression was subsequently destroyed.

8.2 Evidence Evaluation — Updated with New Findings

Evidence Item	Channel	Supports Exfil?	Confidence	Notes
ZIPPER.EXE in unallocated Prefetch	—	YES — Direct	HIGH	File archiver was run; PF deleted — staging confirmed
IMAPI Event 7036 RUNNING/STOPPED 2009-11-17 02:31:53–59	Optical disc	YES — Direct	HIGH	6-second burn cycle confirmed by SysEvent — something burned
IMAPI during HELIX session (05:07:08–15, 2009-11-13)	Optical disc	YES — Indirect	MEDIUM	CD burning active while forensic tool ran
LaCie HDD connected at 02:20:07	USB	YES — Direct	HIGH	External HDD, unknown contents
Fastfat Event 50 at 02:25:28 on HarddiskVolume3	USB	YES — Direct	HIGH	FAT volume error — write issue on removable drive
Definitions.odt accessed 10 min before LaCie	Collection	YES — Indirect	MEDIUM	Document open just before HDD connected
USPTO.gov patent research (16 pages)	Recon	YES — Indirect	MEDIUM	Patent intelligence gathering
Webmail access (login_username='jo' × 5)	Webmail	POSSIBLE	LOW-MEDIUM	Access confirmed; actual email content unknown
Outlook Express (MSIMN.EXE × 4)	Email	POSSIBLE	LOW	Client used; only 2 old test emails recovered
mstsc.exe accessed 2009-11-11	Lateral	POSSIBLE	LOW-MEDIUM	RDP may enable internal staging — not confirmable
No archive files in allocated disk space	—	AGAINST	MEDIUM	No .zip/.rar in allocated — deleted or on external media
Zero keyword search hits (literal/regex)	—	AGAINST	MEDIUM	No plaintext exfil strings — consistent with careful actor

Evidence Item	Channel	Supports Exfil?	Confidence	Notes
HELIX.EXE + EventLog shutdown (65 sec)	Cleanup	YES — Indirect	HIGH	Post-HELIX shutdown = deliberate cover; HELIX used to understand forensic traces
DEFRAG.EXE 2009-11-15	Anti-forensic	YES — Indirect	MEDIUM	Overwrites slack/unallocated — destroys file residue

8.3 Final Exfiltration Determination

[FINAL DETERMINATION — HIGHLY PROBABLE EXFILTRATION] Data exfiltration from M57.biz via Jo's machine is assessed as HIGHLY PROBABLE with MEDIUM-HIGH confidence. Evidence for this determination: (1) ZIPPER.EXE was executed (compression for exfiltration) and its Prefetch was subsequently deleted — demonstrating both data staging and counter-forensic cleanup. (2) IMAPI CD-Burning was confirmed RUNNING and STOPPED on 2009-11-17 02:31:53–59 by the System Event Log — something was burned to optical media on the acquisition day. (3) A LaCie external HDD was connected at 02:20:07, coinciding with Fastfat write warnings on an unrecognised FAT volume. (4) Multiple channels (disc, USB, webmail) were prepared or activated. The specific files transferred cannot be confirmed without the LaCie drive image and optical disc contents.

8.4 What Remains Unconfirmed & Next Steps

1. **Forensic image the LaCie HDD (00D04B881007C255):** Single highest-priority action. Its content would confirm or deny USB exfiltration and identify transferred files.
2. **Retrieve and image the optical disc:** The IMAPI burn on acquisition day produced a disc. Physical recovery of the media from Jo's workstation area is essential.
3. **M57.biz email server logs:** SMTP and webmail server logs would reveal outbound messages from Jo's account during the investigation window.
4. **Full Outlook Express DBX extraction:** Autopsy recovered only 2 old test emails. Manual extraction with DBXtract or Aid4Mail is needed for complete inbox/sent folder review.
5. **Network logs and proxy records:** Any corporate proxy or firewall logs would reveal web-based upload activity from Jo's machine IP address.
6. **RDP server-side logs:** If mstsc.exe was used to connect to other M57.biz servers, those servers' Event Logs would document what Jo accessed.
7. **SYSTEM hive ShimCache parsing:** Deploy AppCompatCacheParser (Eric Zimmerman) against the SYSTEM hive to recover full execution history — would confirm or deny additional tools not captured in Prefetch.

Task 9 — Conclusions & Recommendations

9.1 Summary of All Confirmed Findings

Finding	Confidence	Key Evidence
Machine M57-JO is Jo's Windows XP SP3 workstation at M57.biz	HIGH	OS Info CSV, Registry
ZIPPER.EXE file compression tool was executed (now deleted from Prefetch)	HIGH	Unallocated Prefetch CSV
IMAPI CD-Burning confirmed running/stopped on 2009-11-17 (something burned)	HIGH	sys-event.xlsx Event 7036
LaCie Rugged HDD connected at 02:20:07 on acquisition day	HIGH	USBSTOR Registry
Fastfat Event 50 warning on \Device\HarddiskVolume3 (removable FAT volume)	HIGH	sys-event.xlsx Event 50
HELIX.EXE executed — forensic counter-tool awareness	HIGH	Prefetch CSV
System shutdown 65 seconds after HELIX ran — deliberate pattern	HIGH	Prefetch + SysEvent 6006
MDD_1.3.EXE executed on acquisition day — RAM dump from /RAM path	HIGH	Prefetch CSV
Jo's identity confirmed via webmail form autofill (login_username='jo' × 5)	HIGH	Web_Form_Autofill CSV
Webmail login failed with 'Unknown user or password incorrect' (3 retries)	HIGH	Web_History CSV
USPTO.gov patent office browsed 16 times across two sessions	HIGH	Web_History CSV
Python 2.6.4 and Firefox installed on same day as HELIX activity (2009-11-13)	MEDIUM	AppEvent + Installed Programs
DEFRAG.EXE run 2009-11-15 — disk defragmentation (anti-forensic possible)	MEDIUM	Prefetch CSV
JAVA.EXE executed from anomalous \SYSTEM32 path	MEDIUM	Prefetch CSV
mstsc.exe (Remote Desktop) accessed 2009-11-11 — possible lateral movement	MEDIUM	Remote_Monitoring CSV
No malicious persistence detected in Registry Run keys or services	HIGH	Autopsy Registry analysis
Zero literal/regex keyword hits — no plaintext IOC strings in allocated space	HIGH	TSK_KEYWORD_HIT CSV

9.2 Investigation Limitations

- **LaCie Drive Not Available:** Most critical physical evidence item — not imaged. Exfiltration cannot be fully confirmed without it.
- **Optical Disc Not Recovered:** IMAPI burn confirmed but the disc was not imaged. Burned content is unknown.
- **ShimCache Not Parsed:** SYSTEM hive present but AppCompatCache not extracted — historical execution coverage is partial.
- **Amcache Not Applicable:** Windows XP SP3 — artefact does not exist on this OS version.
- **Security Event Log Cleared:** SecEvent.Evt had no entries post-2009-11-08 — no logon or object access events available.
- **No Network Capture:** Web-based exfiltration (HTTPS upload, webmail) cannot be confirmed from disk alone.
- **OE Mail Not Fully Extracted:** Outlook Express DBX stores require dedicated tools beyond Autopsy's email module.

9.3 Immediate Recommendations (Incident Response)

Priority	Action	Owner
P1 — CRITICAL	Seize and forensically image the LaCie HDD (serial: 00D04B881007C255)	DFIR Team / Legal
P1 — CRITICAL	Recover the optical disc burned on 2009-11-17 02:31:53 from Jo's physical workspace	DFIR Team
P1 — CRITICAL	Preserve M57.biz email server logs (SMTP, webmail) for Jo's account — full window	IT / Legal
P1 — CRITICAL	Issue litigation hold — all digital evidence, communications, and access logs	Legal Counsel
P1 — CRITICAL	Revoke Jo's network and system credentials immediately	IT Security
P2 — HIGH	Interview Jo under caution — address LaCie, HELIX, optical disc, USPTO browsing	HR / Legal
P2 — HIGH	Parse SYSTEM hive ShimCache using AppCompatCacheParser (Eric Zimmerman tools)	DFIR Team
P2 — HIGH	Manually extract Outlook Express DBX mail stores using DBXtract or Aid4Mail	DFIR Team
P2 — HIGH	Review M57.biz RDP server logs for connections from M57-JO on 2009-11-11	Security Ops
P2 — HIGH	Examine \Device\HarddiskVolume3 — determine if this maps to LaCie or another volume	DFIR Team
P3 — MEDIUM	Deploy DLP controls — block large outbound attachments and USB write operations	IT Security
P3 — MEDIUM	Implement USB device whitelist policy (hardware ID enforcement)	IT / Endpoint
P3 — MEDIUM	Deploy SIEM with insider threat rules: bulk file access, USB, CD burning, forensic tools	Security Ops

Priority	Action	Owner
P3 — MEDIUM	Conduct patent document access audit — all employees — 90-day lookback	Security Ops

9.4 Long-Term Security Recommendations

8. Implement a formal Data Classification Policy for all patent and trade secret documents, with DRM controls and access auditing.
9. Deploy User and Entity Behaviour Analytics (UEBA) to detect anomalous bulk file access, large email attachments, USB write operations, and off-hours activity.
10. Enforce least-privilege access — employees should only access patent documents directly relevant to their current assignment.
11. Require all removable media to be registered, encrypted, and audited before authorised connection.
12. Block or alert on CD/DVD burning service (IMAPI) activation except when explicitly authorised.
13. Monitor and alert on execution of forensic tools (HELIX, MDD, WinHex, etc.) on corporate endpoints via application whitelisting.
14. Establish a documented Insider Threat Detection programme with behavioural baselines, HR integration, and defined escalation procedures.
15. Maintain centralised, tamper-evident SIEM logging (including USB plug/unplug events, service starts/stops, and process execution) for all endpoints.

Appendices

Appendix A — Tools & Versions

Tool	Version	Purpose
FTK Imager (Exterro)	4.7.3.81	Image mounting, hash verification (MD5 + SHA1), E01 validation
Autopsy	4.23.0	Primary analysis — all ingest modules, CSV exports, file system parsing
Velociraptor	0.6.8	Threat hunting simulation — VQL methodology applied to CSV evidence
Microsoft Excel / OpenPyXL	Current	Event Log XLSX parsing (sys-event.xlsx, app-event.xlsx)
Python 3	Current	CSV parsing and analysis of Autopsy exports

Appendix B — Evidence Files Analysed

File	Rows	Key Findings
Operating_System_Information CSV	1	WinXP SP3, M57-JO, Owner: Jo, M57.biz
USB_Device_Attached CSV	6	LaCie Rugged HDD — CRITICAL
Run_Programs CSV (Prefetch)	49	HELIX, MDD, CMD, DEFRAG, IMAPI × 14, WMIC
Prefetch_20260429123907 CSV	86	Includes ZIPPER.EXE in unallocated — CRITICAL
sys-event.xlsx	169	IMAPI confirmed, Fastfat Event 50, HELIX shutdown, all boots
app-event.xlsx	91	Python + Java + WebFldrs install confirmations
Web_History CSV	3,557	USPTO.gov, webmail.m57.biz, failed login, personal browsing
Web_Search CSV	36	Firefox/python downloads, personal; no exfil searches
Web_Form_Autofill CSV	8	login_username='jo' × 5 — CONFIRMS Jo's identity at webmail
Web_Downloads CSV	3	python-2.6.4.msi, Firefox setup, R79733 (AVG)
Web_Bookmarks CSV	96	Default FF + personal bookmarks
Shell_Bags CSV	25	Desktop shortcuts, folder navigation history
Recent_Documents CSV	2	Definitions.odt, devmgmt.msc
Installed_Programs CSV	95	Python, Firefox, OpenOffice, AVG, Java, 40+ patches
Default_Default CSV (Email)	2	Old test messages — 2001, 2004 — not investigation-relevant

File	Rows	Key Findings
config CSVs (Registry hives)	25	Confirms all hives present; SYSTEM last modified 02:22:31
Remote_Monitoring CSV	4	mstsc.exe, Atera log.txt, Kaseya system.LOG flagged
Suspicious_Items CSV	1,342	Primarily OO gallery / Java cache YARA false positives
TSK_KEYWORD_HIT CSV	3	Literal=0, Regex=0, Email=1 (3,515 matches) — interpreted in Task 3.2

Appendix C — Hash Values

Item	Value
Image MD5	f3160a776ca1d59172c6acb622b67459
Image SHA1	cf2d8ca1e282501dc7977acd4ac48df1a4b23e9f
Sector Count	25,429,824
LaCie Device ID	00D04B881007C255

Appendix D — MITRE ATT&CK Reference

ID	Technique	Observed Evidence	Status
T1005	Data from Local System	Definitions.odt + patent research	POSSIBLE
T1021.001	Remote Services: RDP	mstsc.exe accessed 2009-11-11	POSSIBLE
T1047	WMI Execution	WMIC.EXE × 5 (2009-11-16 20:47)	CONFIRMED
T1052	Exfiltration via Physical Medium	IMAPI × 14; confirmed RUNNING 2009-11-17	CONFIRMED
T1052.001	Exfiltration via USB	LaCie HDD connected; Fastfat Event 50	CONFIRMED
T1059.003	Command Shell	CMD.EXE × 6 (last: 2009-11-17 02:21:47)	CONFIRMED
T1059.006	Python	Python 2.6.4 installed; runtime available	CONFIRMED
T1070	Indicator Removal	HELIX.EXE + EventLog shutdown (65 sec)	CONFIRMED
T1070.004	File Deletion	ZIPPER.EXE Prefetch deleted; DEFRAG	CONFIRMED
T1110	Brute Force	3× failed webmail login 21:35–21:36	POSSIBLE
T1114	Email Collection	Webmail + Outlook Express access	POSSIBLE

ID	Technique	Observed Evidence	Status
T1560	Archive Collected Data	ZIPPER.EXE (deleted Prefetch)	CONFIRMED
T1588.002	Obtain Tool: Forensic Tools	HELIX.EXE + MDD_1.3.EXE	CONFIRMED
T1593	Search Open Technical Databases	USPTO.gov × 16 visits	CONFIRMED

— END OF REPORT — FINAL EDITION —

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